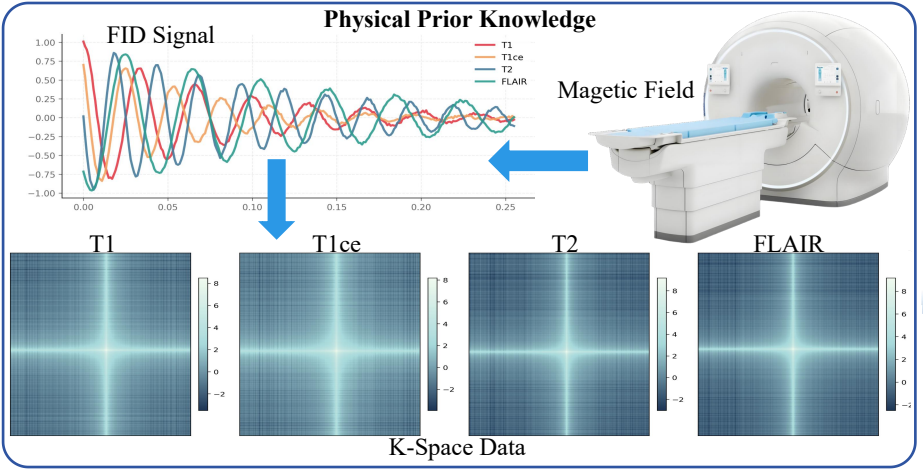
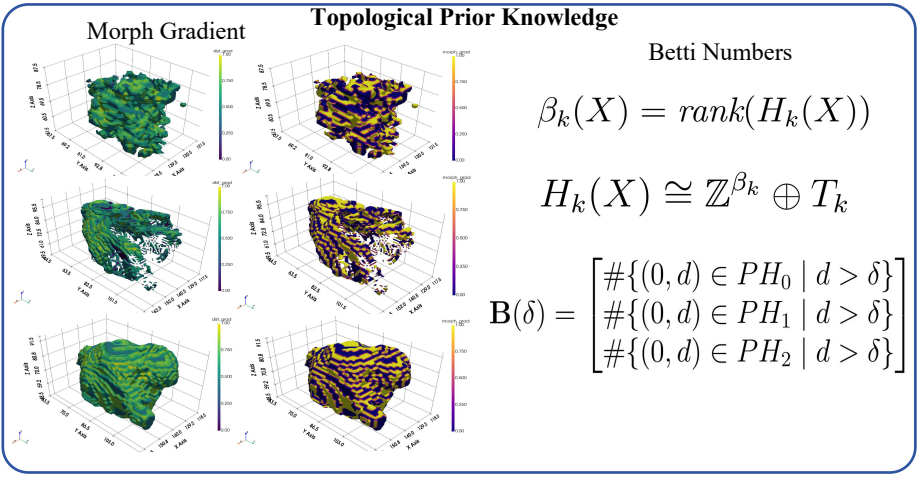
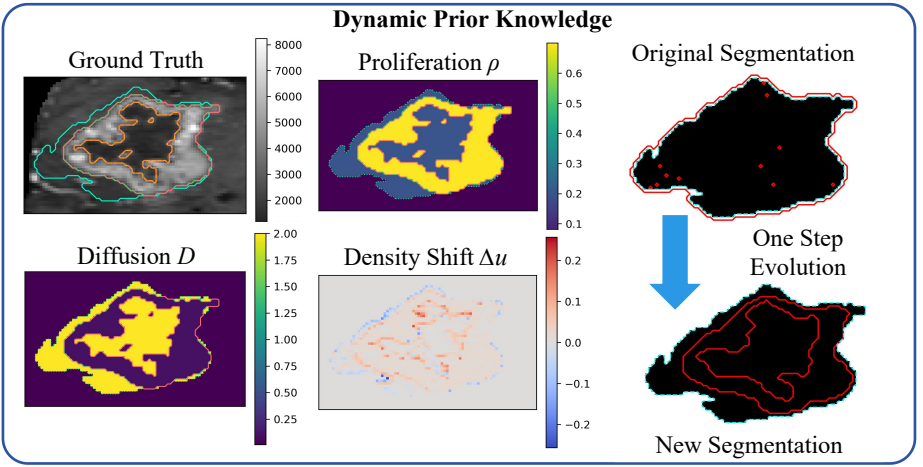
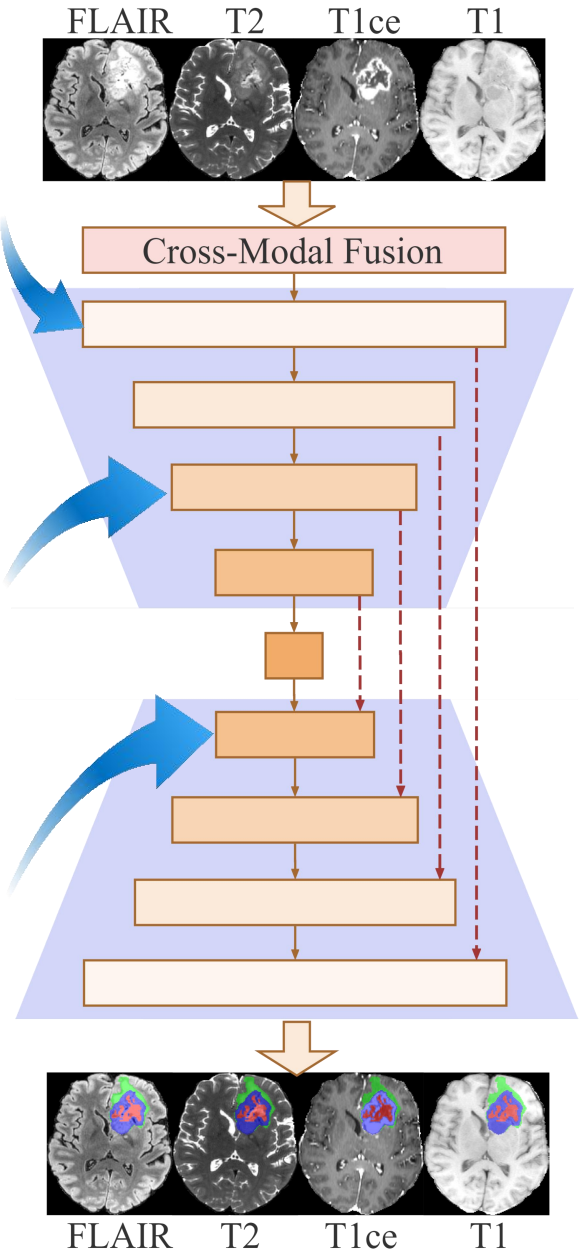

Physical Prior Knowledge
FID Signal
T1
T1ce
T2
FLAIR
Magetic Field
T1
T1ce
T2
FLAIR
K-Space Data


Topological Prior Knowledge
Morph Gradient
Betti Numbers
$\beta_k(X) = rank(H_k(X))$
$H_k(X) \cong \mathbb{Z}^{\beta_k} \oplus T_k$
$\mathbf{B}(\delta) = \begin{bmatrix} \#\{(0,d) \in PH_0 \mid d > \delta\} \\ \#\{(0,d) \in PH_1 \mid d > \delta\} \\ \#\{(0,d) \in PH_2 \mid d > \delta\} \end{bmatrix}$


Dynamic Prior Knowledge
Ground Truth
Proliferation ρ
Original Segmentation
Diffusion D
Density Shift Δu
One Step Evolution
New Segmentation


FLAIR
T2
T1ce
T1
Cross-Modal Fusion
FLAIR
T2
T1ce
T1